

MANUFACTURING PROCESS-I*Time Allowed: 3 Hours**Full Marks: 60***Answer the following questions from Group-A, B & C as directed.****Group A**

1. Choose the correct answer from the alternatives (any ten):

1 x 10

- i. An example of resistance welding is - a) TIG, b) MIG, c) Submerged arc welding, d) Spot welding.
- ii. Non-Consumable electrode used in- a) GMAW, b) GTAW, c) MMAW, d) Thermit.
- iii. A welding defect is - a) crack, b) crash, c) scratch, d) swell.
- iv. In which of the following processes, material is neither added nor removed but is deformed into desired shape?- (a) Surface finishing process, (b) Metal forming process, (c) Casting, (d) Machining.
- v. Mechanical property desired for successful forging is - a) Ductility, b) Malleability, c) Hardness, d) Toughness.
- vi. The most commonly used flame in gas welding is - a) Neutral b) Oxidizing c) Carburizing d) All of the above.
- vii. Structural section such as rails, angles, I beam are made by - a) Hot extrusion, b) Hot rolling, c) Hot drawing, d) Hot spinning.
- viii. Half nut is connected with - a) Milling machine, b) Locking device, c) Jigs & Fixture, d) Thread cutting on Lathe.
- ix. Long wires are made by following process - a) Extrusion, b) Rolling, c) Piercing, d) Drawing.
- x. A moving mandrel is used in - (a) wire drawing, (b) tube drawing, (c) metal cutting, (d) casting.
- xi. Which of the following flame is harmful to steel? - (a) Neutral flame, (b) Oxidizing flame, (c) Carburizing flame, (d) Both oxidizing flame and carburizing flame.
- xii. Which of the following materials is used as the pattern material in precision investment casting? a) Metal, b) Molten wax, c) Plastics, d) Wood.
- xiii. Blanking and punching operations can be performed simultaneously on – (a) compound die, (b) combination die, (c) progressive die, (d) simple die.
- xiv. A casting defect which results in general enlargement of a casting is known as- (a) Shift, (b) sand wash, (c) swell, (d) scab.
- xv. Core print is a part of - a) core, b) pattern, c) core box, d) chaplet.

2. Fill in the blanks (any ten):

1x10

- i. Extra metal which settles down in the gutter is known as _____.
- ii. The best shape for sprue in sand-casting is _____.
- iii. Voltage required for Arc welding is _____ volts.

- iv. Oxygen cylinder used in oxy-acetylene gas welding is usually painted in _____ colour.
- v. In sand moulding, the middle part of the flask is called _____.
- vi. A _____ is a tool used for enlarging a previously drilled hole.
- vii. In blanking operation, clearance is provided on _____.
- viii. Collets, angle plates and mandrel are _____ of a lathe.
- ix. A small depression in the parent metal during arc welding is called _____.
- x. Lathe is specified by _____ of the job over carriage.
- xi. Large size bolt heads are made by _____ forging.
- xii. Model of an object which is used for making mould in sand moulding is called _____.
- xiii. Collapsible paste tubes are produced by _____.
- xiv. In _____ process, a specified shape of metal from the side or edge of a metal sheet can be removed using suitable punch and die.
- xv. _____ is a mixture of finely divided aluminium and iron oxide.

3. Answer the following questions (any ten):

1x10

- i. What is a chaplet?
- ii. Why electrodes are coated in manual arc welding?
- iii. State the relationship between lead and pitch of a thread?
- iv. Sketch three types of gas welding flames.
- v. Which sheet metal operation is used to make coins?
- vi. What is used to reduce friction during machining operation?
- vii. Define orthogonal cutting.
- viii. Pressurised gating system is used for casting of which metals and alloys?
- ix. Write down Taylor tool life equation?
- x. What is the use of filler metal in welding?
- xi. What is core print?
- xii. Which manufacturing process is used for manufacturing of Crankshaft?
- xiii. Define tool life for a single point HSS cutting tool in lathe operation.
- xiv. Which welding is used for sheet metal?
- xv. In which furnace cast iron is melted?

Group B

4. Answer the following questions (any six):

2x6

- i. What is the difference between brazing and soldering?
- ii. State the purpose of using cutting fluids in machining operations.
- iii. Name different defects of rolling process.
- iv. Give examples of single point cutting tool & multi point cutting tool.
- v. State specification of a Centre lathe.
- vi. Write two advantages of submerged arc welding.
- vii. What is Carburising flame?
- viii. Differentiate between hot working process and cold working process.
- ix. What are the components of a carriage?
- x. What is swell?

Group C

5. Answer the following question (any one):

1x6

- i. How manufacturing processes are classified? Explain with example.
- ii. Explain hydrostatic extrusion process with a simple diagram.
- iii. Briefly explain the processes - a) cutting, b) lancing, c) embossing.

5+1

6

2+2+2

6. Answer the following question (any one): 1x6
- i. Write down the favourable conditions for continuous and discontinuous chip formation? 6
 - ii. Write short notes on – a) Facing, b) Parting-off, c) Knurling. 2+2+2
 - iii. What are the Desirable properties of cutting tool materials and cutting fluid? 3 + 3
7. Answer the following question (any one): 1x6
- i. Describe the processes TIG and MIG briefly. 3+3
 - ii. Describe submerged arc welding process with sketch. 6
 - iii. Name the various types of defects that occur in sand castings. State causes and remedies in brief. 2+4
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